

Information Technology University of the Punjab
SE460T Cloud Computing with Big Data Applications – Fall 2025
Quiz 4

Name: _____

Roll No.: _____

Time allowed: _____

Maximum Marks: 16

1. What is the primary purpose of HDFS?
 - a. Low-latency transaction processing
 - b. Distributed file storage for massive datasets
 - c. Real-time stream processing
 - d. In-memory caching
2. Which component of HDFS manages metadata such as file names, block locations, and permissions?
 - a. DataNode
 - b. ResourceManager
 - c. NameNode
 - d. NodeManager
3. What is the default replication factor in HDFS?
 - a. 1
 - b. 2
 - c. 3
 - d. 5
4. Which of the following best describes the primary role of YARN in the Hadoop ecosystem?
 - a. Distributed storage of large datasets
 - b. Cluster-wide resource management and job scheduling
 - c. Data serialization and compression
 - d. Real-time stream processing
5. Which YARN component is responsible for accepting job submissions and launching the first container for an application?
 - a. Scheduler
 - b. Node Manager
 - c. Application Manager
 - d. Application Leader
6. If a task fails, the Application Leader is responsible for detecting the failure and requesting new containers to retry it.
 - a. True
 - b. False
7. In MapReduce, application developers are responsible for task scheduling and data distribution across the cluster.
 - a. True
 - b. False
8. Which statement best captures the main value proposition of the MapReduce programming model?
 - a. It provides low-latency access to transactional data
 - b. It abstracts distributed execution while requiring users to manage fault tolerance
 - c. It allows users to define Map and Reduce logic while the framework manages parallelism and failures
 - d. It is primarily designed for real-time stream processing
9. If a worker node fails during execution, MapReduce redistributes its tasks to other nodes automatically.
 - a. True
 - b. False
10. Which of the following is a valid output of a Map function in a word count application?
 - a. (document, total_words)
 - b. (word, 1)
 - c. (word, total_count)
 - d. (file, frequency_table)

11. Which phase of MapReduce is responsible for transferring and grouping mapper output so that all values for the same key reach the same reducer?
- Mapper phase
 - Partitioning phase
 - Shuffle & Sort phase
 - Reducer phase
12. Apache Flink supports both batch processing and streaming processing through different APIs.
- True
 - False
13. Which statement correctly describes Presto?
- It executes SQL queries using MapReduce jobs
 - It is optimized for batch ETL workflows
 - It is an in-memory distributed SQL engine optimized for interactive queries
 - It is a scripting language for data transformation
14. Apache Spark was primarily designed to overcome which major limitation of the MapReduce model?
- Lack of SQL support
 - Disk-based processing between stages causing high latency
 - Inability to scale across clusters
 - Absence of fault tolerance
15. Which Spark module provides low-latency, interactive SQL queries with optimization features such as columnar storage and code generation?
- Spark Core
 - Spark Streaming
 - Spark GraphX
 - Spark SQL
16. Which Amazon EMR node type is responsible for cluster coordination and can exist as the only node in a single-node cluster?
- Core node
 - Task node
 - Main node
 - Worker node